

Mechanics HW

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Question 1

(a)

Consider a system in which the apparent gravity is pointing down, (the force MA from the centre of mass)
Then when the door begins to swing it converts its gravitational potential energy to rotational kinetic energy
After swinging through an angle of $\frac{\pi}{2}$, the centre of mass of the door has fallen a distance of $\frac{w}{2}$

$$MA \left(\frac{w}{2} \right) = \frac{1}{2} I_0 \omega^2$$

Note: the moment of inertia of the door is $\frac{1}{3} M w^2$

$$\Rightarrow \omega = \sqrt{\frac{3A}{w}}$$

(b)

At the point at which the door has swung through an angle of $\frac{\pi}{2}$, the apparent gravity is point straight down the door, also there is a centrifugal force required

Let F_u be the force pointing upwards in the frame with the apparent gravity pointing down

Then

$$\begin{aligned} F_u &= MA + M\omega^2 \left(\frac{w}{2} \right) \\ \Rightarrow F_u &= MA + \frac{3MA}{2} = \frac{5MA}{2} \end{aligned}$$

Question 2

(a)

$$\begin{aligned} U(\infty) - U(r) &= - \int_r^\infty - \frac{2A}{r'^3} dr' \\ &= \frac{A}{r^2} \end{aligned}$$

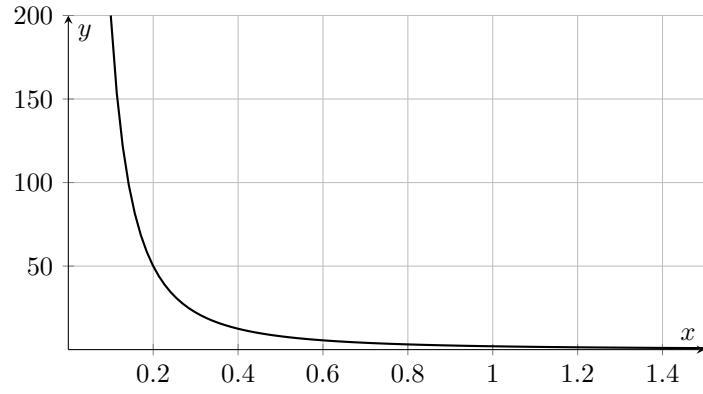
Taking $U(\infty) = 0$, we get

$$U(r) = -\frac{A}{r^2}$$

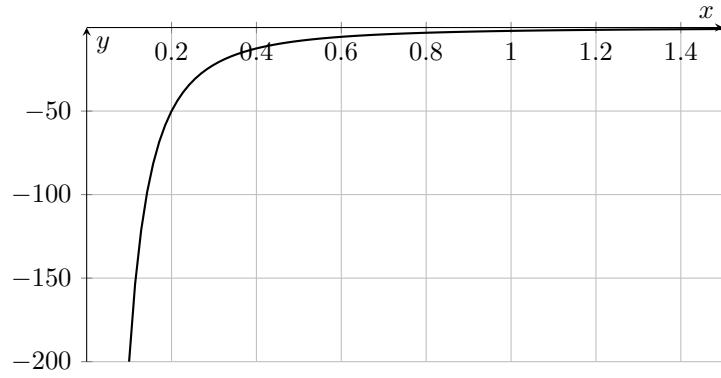
Then

$$U_{\text{eff}}(r) = \frac{l^2}{2mr^2} - \frac{A}{r^2} = \frac{1}{r^2} \left(\frac{l^2}{2m} - A \right)$$

When $l^2 > 2mA$



When $l^2 < 2mA$



When $l^2 = 2mA$, $U_{\text{eff}}(r) = 0 \forall r$

(b)

Uniform radial velocity i.e. $\ddot{r} = 0$

$$\begin{aligned}
 ma_r &= F(r) \\
 m(\ddot{r} - r\dot{\theta}^2) &= -\frac{2A}{r^3} \\
 \ddot{r} &= r\dot{\theta}^2 - \frac{2A}{mr^3} \\
 &= r \left(\frac{l^2}{m^2 r^4} \right) - \frac{2A}{mr^3} \\
 &= 0 \\
 \Rightarrow l^2 &= 2mA \\
 \Rightarrow l &= \sqrt{2mA}
 \end{aligned}$$

(c)

$$\begin{aligned}
 \dot{\theta} &= \frac{l}{mr^2} = \frac{\sqrt{2mA}}{mr^2} \\
 \dot{\theta} &= \frac{d\theta}{dr} \frac{dr}{dt} \\
 \Rightarrow \frac{d\theta}{dr} &= \frac{\sqrt{2mA}}{mvr^2} \\
 \Rightarrow \theta &= \frac{\sqrt{2mA}}{mv} \int \frac{1}{r^2} dr \\
 \Rightarrow \theta &= -\frac{\sqrt{2mA}}{mvr} + \theta_0
 \end{aligned}$$

Question 3

$$\ddot{I} + 6\dot{I} + 8I = Ce^{-t}$$

(a)

Guess $I(t) = e^{\lambda t}$

$$\Rightarrow (\lambda^2 + 6\lambda + 8)e^{\lambda t} = 0$$

Since $e^{\lambda} \neq 0, \forall t$

$$\Rightarrow \lambda^2 + 6\lambda + 8 = 0$$

Thus $\lambda = -4$ or $\lambda = -2$

$$\Rightarrow I(t) = Ae^{-4t} + Be^{-2t}$$

(b)

Guess particular solution $I(t) = \alpha e^{-t}$

$$\Rightarrow \alpha e^{-t} - 6\alpha e^{-t} + 8\alpha e^{-t} = 6e^{-t}$$

$$\Rightarrow \alpha = 2$$

Thus the particular solution is $I(t) = 2e^{-t}$

(c)

To find the general solution, we simply add the general solution and the particular solution together

$$\Rightarrow I(t) = Ae^{-4t} + Be^{-2t} + 2e^{-t}$$

Question 4

First I will compute the moment of inertia of a ring with outer radius r_2 and inner radius r_1

$$I_0 = \int_{\text{object}} r^2 dm$$

$$dm = \lambda r dr d\theta, \quad \lambda = \frac{M}{\pi(r_2^2 - r_1^2)}$$

$$\begin{aligned} I_0 &= \frac{M}{\pi(r_2^2 - r_1^2)} \int_0^{2\pi} \int_{r_1}^{r_2} r^3 dr d\theta \\ &= \frac{M}{2(r_2^2 - r_1^2)} (r_2^4 - r_1^4) \\ &= \frac{M}{2} (r_2^2 + r_1^2) \end{aligned}$$

(a)

For a circle $r_2 = r_1 = R$

$$\Rightarrow I_0 = MR$$

(b)

For a solid disk $r_1 = 0, r_2 = R$

$$\Rightarrow I_0 = \frac{MR}{2}$$